KAZAKH BRITISH TECHNICAL UNIVERSITY

WEB DEVELOPMENT

Lesson 8

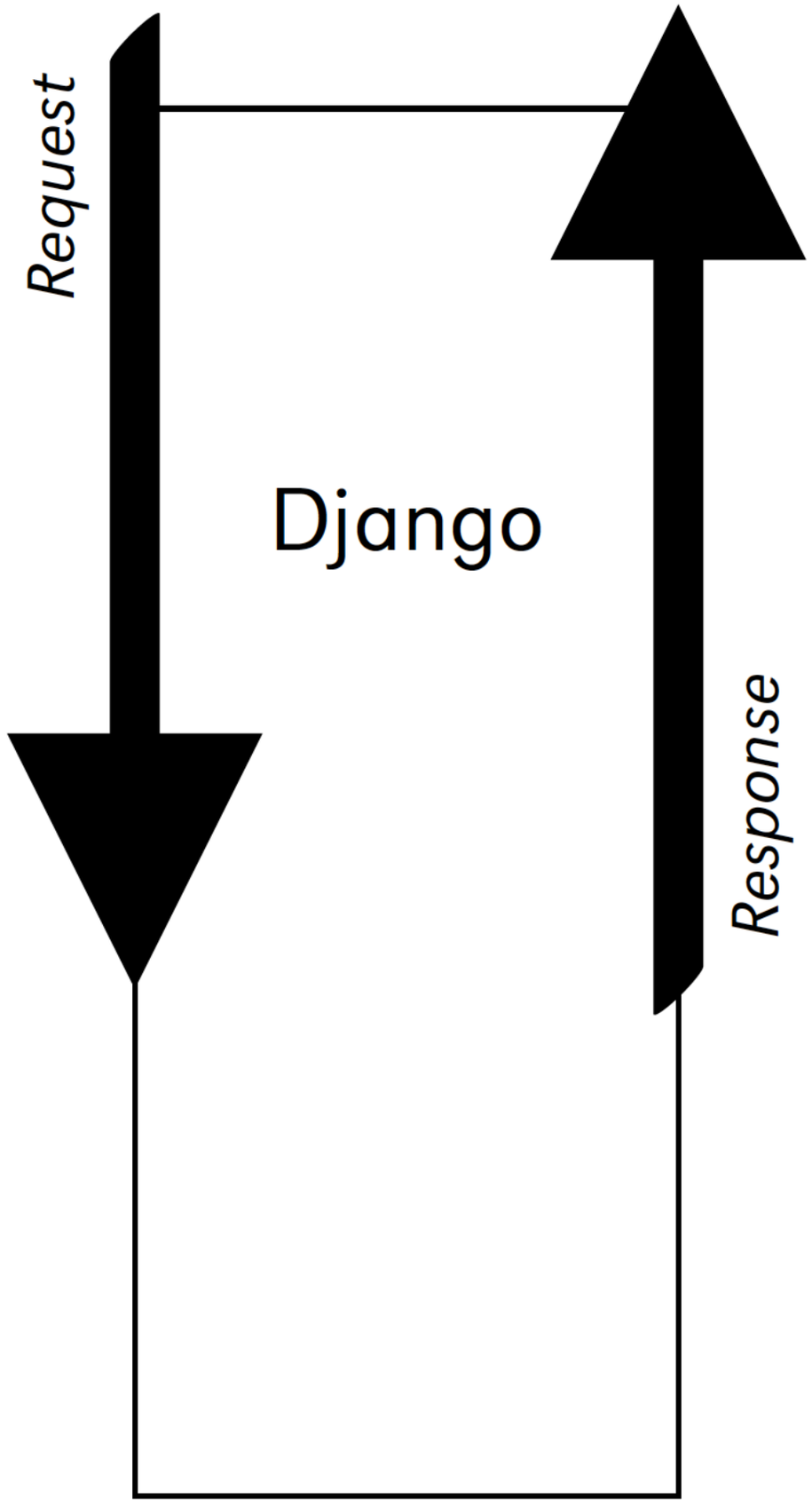
What is Django?

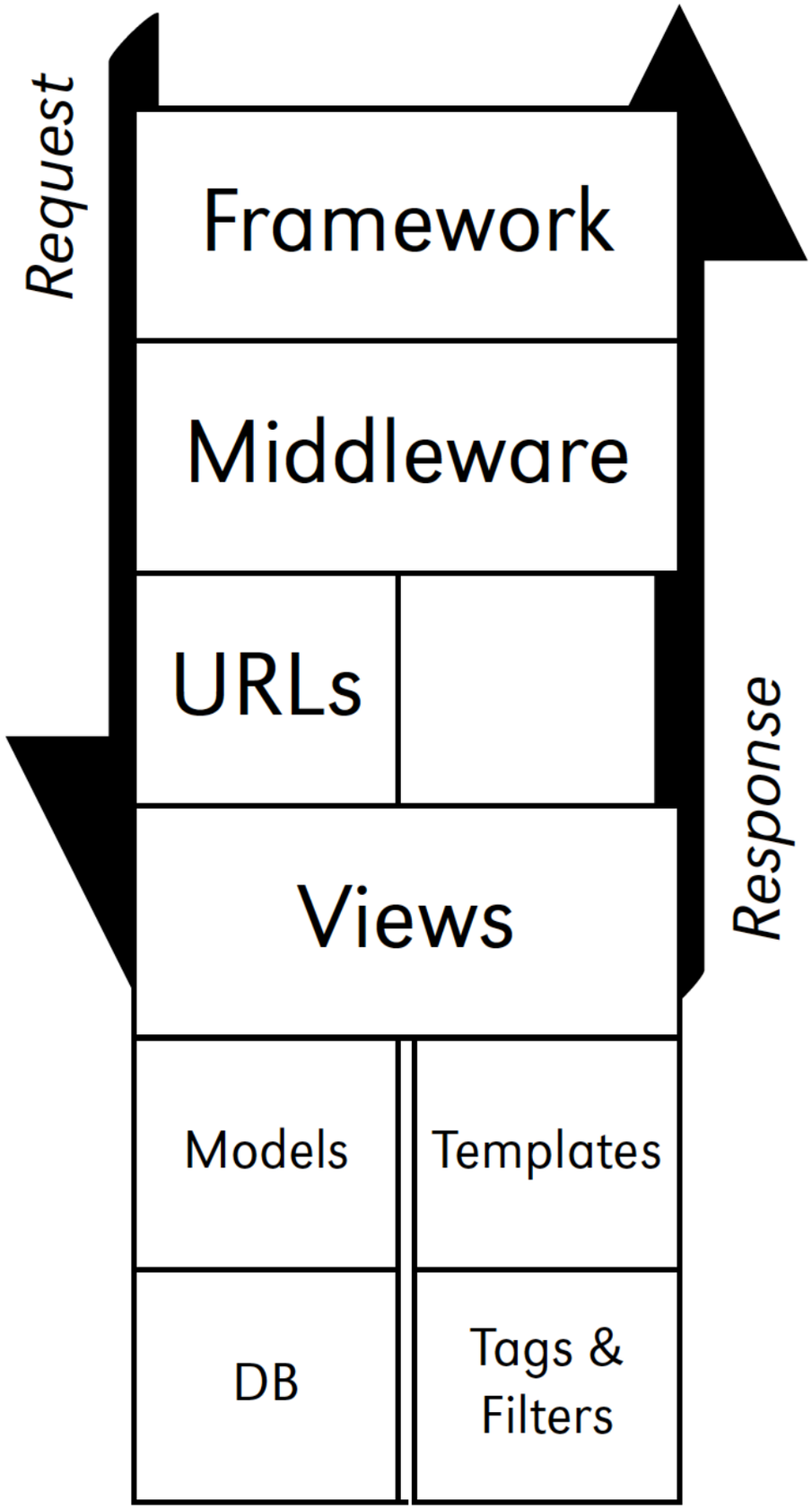
What is Django?

- High-level framework for rapid web development
- Complete stack of tools
- Data modelled with Python classes
- Production-ready data admin interface, generated dynamically
- Elegant system for mapping URLs to Python code
- Generic views' to handle common requests

Django Components

- Think MTV instead of MVC
- Models - Django ORM
- Templates - Django Template Engine
- Views - Python function, Request in Response out
- URL Patterns - Regular expression based


Request
Django
Response


Request
Framework
Middleware
URLs
Views
Response
Models
Templates
DB
Tags &
Filters

Defining Requirements

- requirements.txt

```
# Create requirements.txt for current env
$ pip freeze > requirements.txt

# Install all modules from requirements.txt file recursive
$ pip install -r requirements.txt
```

Starting a Project

```
# Mac/Linux/etc...

$ pip install django
$ django-admin startproject demo
$ cd demo
$ python manage.py migrate
$ python manage.py runserver

# Windows

> pip install django
> python Scripts/django-admin.py startproject demo
> cd demo
> python manage.py migrate
> python manage.py runserver
```

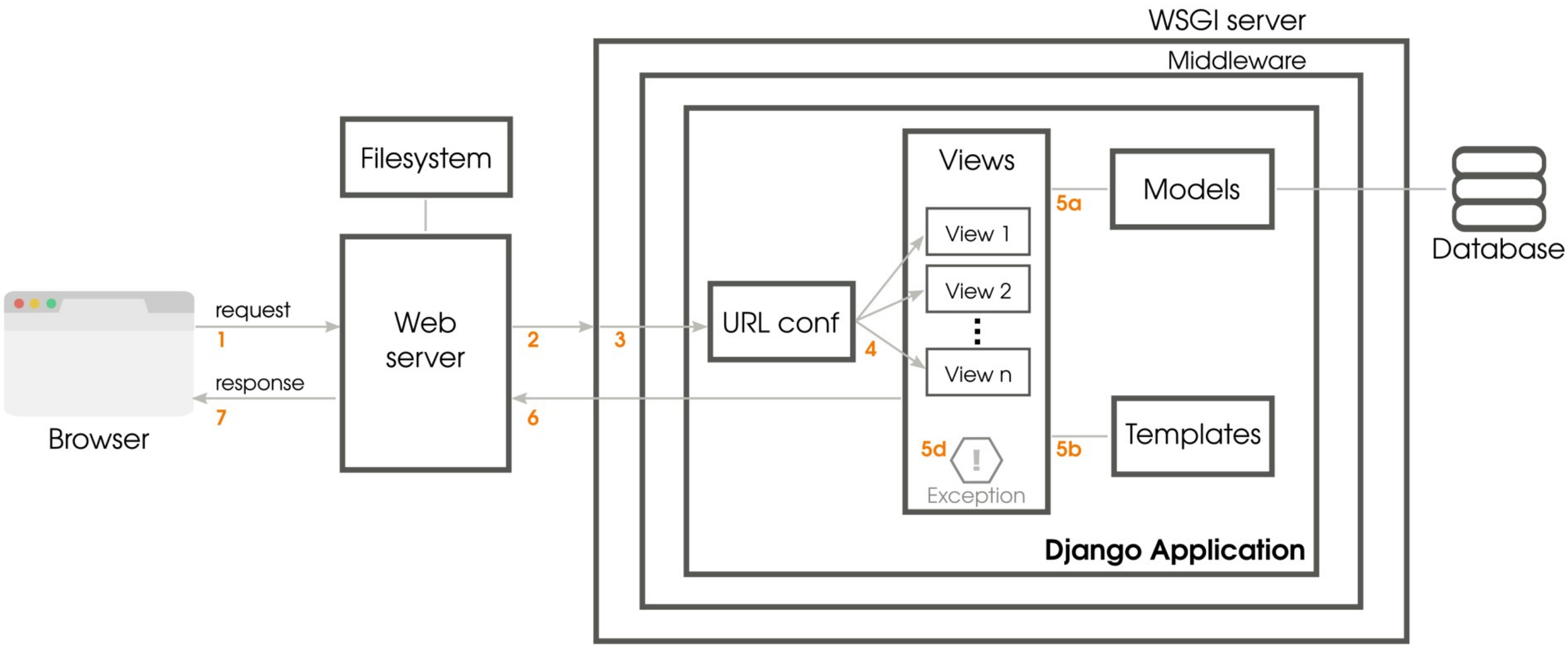
URLs

- Map URLs in requests to code that can be executed
- Regular expressions!
- Subsections of your site can have their own urls.py modules

Views

- Code that handles requests
- Other frameworks often call these “controllers”
- Basically a function that:
  - gets a request passed to it
  - returns text or a response

How Does Django Works?



Regular Expressions in URL

`.` — (dot) Any character

`\d` — Any digit

`[A-Z]` — Any character A-Z (uppercase)

`[a-z]` — Any character, a-z (lowercase)

`[A-Za-z]` — Any character, a-z (case insensitive)

`+` — One or more of the previous expression

`[^/]+` — All characters except forward slash

`?` — Zero or more of the previous expression

`{1,3}` — Between one and three (inclusive) of the previous expression

How Django Processes a Request

- `python manage.py runserver`
  - `settings.py`
  - `ROOT_URLCONF`
- `/time/`
  - look through all urls patterns and compare
    - `HTTPRequest` object as first parameter
    - view function responsible to return `HTTPResponse`

404 Errors

Word About Pretty URLs

```
re_path(r'^time/plus/\d+/$', hours_ahead)
```

- /time/plus/1/
- /time/plus/2/
- /time/plus/3/
- /time/plus/1000000/

Word About Pretty URLs

```
re_path(r'^time/plus/\d{1,2}/$', hours_ahead)
```

- /time/plus/1/
- /time/plus/2/
- /time/plus/3/
- /time/plus/1000000/

Passing that data to the view function

parentheses around the data

```
re_path(r'^time/plus/(\d{1,2})/$', hours_ahead)
```

Coding Order

1. views —> urls

2. urls —> views

Django new App

The Django Template System

template might be as HTML file

render function

MTV

- **`M (Model)`** — data access layer
- **`T (Template)`** — presentation layer
- **`V (View)`** — controller layer

- ***`S (service layer)`*** — *business logic layer*

Configuring the Database

Configuring the Database

settings.py

- `DATABASE_ENGINE = ''`
- `DATABASE_NAME = ''`
- `DATABASE_USER = ''`
- `DATABASE_PASSWORD = ''`
- `DATABASE_HOST = ''`
- `DATABASE_PORT = ''`

Configuring the Database

`settings.py`

- `DATABASE_ENGINE = ''`

Settings	Database	Required Adapter
`postgresql`	PostgreSQL	`psycopg` version 1.x
`postgresql_psycopg2`	PostgreSQL	`psycopg` version 2.x,
`mysql`	MySQL	`MySQLdb`
`sqlite3`	SQLite	`pysqlite`
`ado_mssql`	Microsoft SQL Server	`adodbapi` version 2.0.1+
`oracle`	Oracle	`cx_Oracle, oracledb`

Defining Models in Python

Python model → SQL `CREATE TABLE`

Create new migration file if necessary

```
>>> python manage.py makemigrations
```

Execute created migration file

(change database schemas)

```
>>> python manage.py migrate
```

Inserting and Updating Data

- Create instance of model
  - save() // INSERT INTO “table_name” …
- Find object and update class fields
  - save() // UPDATE “table_name” SET ...

Selecting Objects

```
>>> products = Product.objects.all()
```

```
SELECT * FROM products;
```

Filtering Data

```
>>> p = Product.objects.filter(name="Test name")
```

```
SELECT * FROM products
WHERE name='Test name';
```

Filtering Data

- contains / icontains
- startswith / endswith
- istartswith / iendswith
- exact / iexact
- in
- isnull
- gt / gte
- lt / lte

```
>>> p = Product.objects.filter(name__contains="Test")
```

Retrieving Single Objects

```
>>> p = Product.objects.get(id=3)
```

```
SELECT * FROM product WHERE id=3;
```

Ordering Data

```
>>> p = Product.objects.order_by(“name”)
```

```
SELECT * FROM products ORDER BY name;
```

Ordering Data

```
>>> p = Product.objects.order_by(“name”, “address”)

>>> p = Product.objects.order_by(“-name”)
```

Meta class

```
class Product(models.Model):
    ...
    class Meta:
        ordering = [“name”]
```

Chaining Lookups

```
>>> p = Product.objects.filter(name=“Laptop”).order_by(“-price”)
```

Deleting Objects

```
>>> p = Product.objects.get(id=3)
>>> p.delete()

>>> products = Product.objects.all()
>>> products.delete()
```

Questions?